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Day 3: RNNs, LSTM, GRU
Summary

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Sequential Data

Sequence data is type of data with dependent elements and ordered by time

- Many to Many
  - Formation: Sequence in input and sequence in output
  - Example: Machine Translation

- One to Many
  - Formation: One input and sequence in output
  - Example: Music Generation

- Many to One
  - Formation: Sequence in input and one output
  - Example: Sentiment Classification

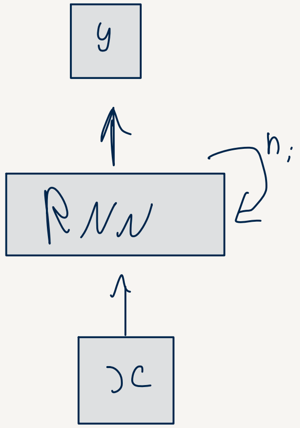
Traditional frameworks faile while processing sequential data because:
1- sequential data varies in the length of the input and the output
2- These networks doesn't share features learned across different positions of text
3- too many parameters

Solution:
- Add Memory (RNNs ...)
- Process the whole sequence as one unit(Transformers ...)

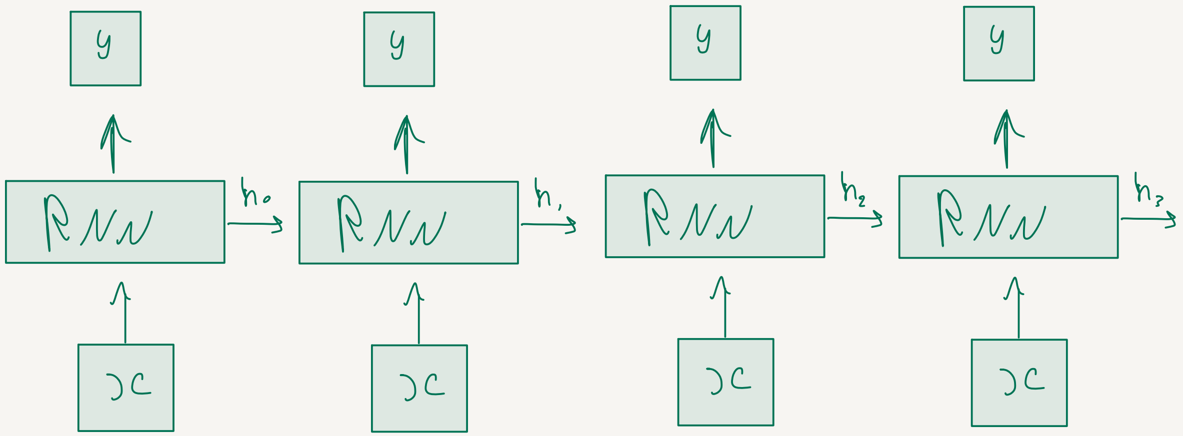
Recurrent Neural Networks (RNNs)

RNN solve sequential dependency problems by adding an internal state from one step to the next time step acting as the network's continuous memory loop

- Folded Representation



- UnFolded Representation



Notice, the rnn cell here is the same with the exact same parameters, it is just used across many time steps

Advantages and Disadvantages

- Advantages

- Can process any length input
- Uses past information
- Model size doesn't increase with longer inputs
- Same weights are applied on every time step

- Disadvantages

- Recurrent computation is slow
- Difficult to access information from many timesteps back
- can't be parallelized

Vanishing Gradient problem: the earlier layers (earlier timesteps) receive very or small gradient updates, so the model tends to forget the earlier inputs

Solution: Gated Architectures

Gated Architectures

Gating is the process of using gates in order to choose between a choice and an alternative choice using a key. Think of it as a water valve that turns on to pass water or blocks it.

Type	States	Gates	How it works
• LSTM	Has one hidden state One cell state	Forget gate Input gate Output gate	The forget gate removes the information that is no longer needed from the memory, the input gate decides what are the new information to store in the cell memory while the output gate decides what information to use at the current step. The cell state works as the long term memory while the hidden state works as the short memory, it is the output at the current step
• GRU	Simpler than LSTM, it has only one memory (hidden state)	Combined input and forget gates in one gate called update gate Has a reset gate	The reset gate decides what information to remove from the hidden state, the update gate decides what information pick from old hidden state and what I for to pick from the new information